TBot-SA$_1$: a 3D-Causal World-Spatial-Action Model for Generalizable Robot Control

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Project Page: `https://zaleni.github.io/TBot-SA1/`

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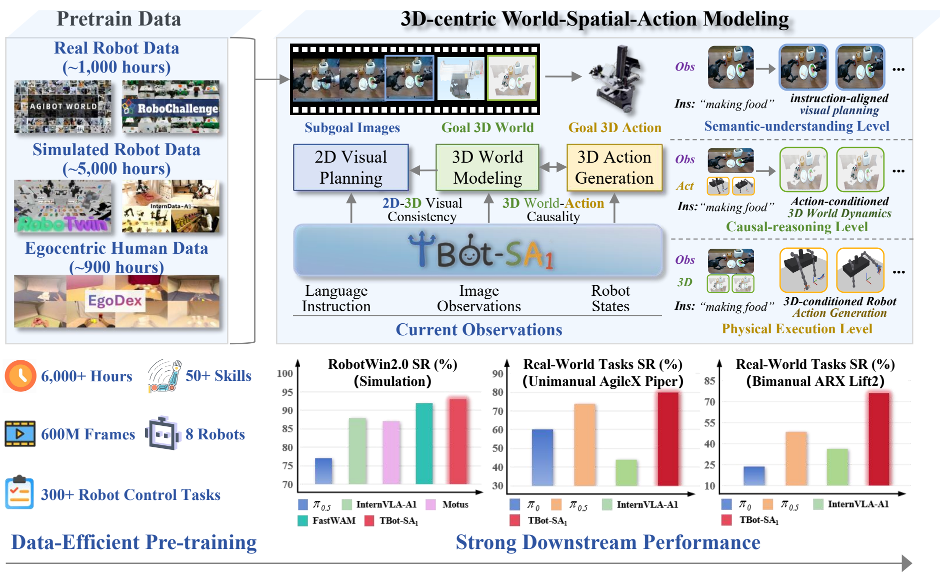


Figure 1: **TBot-SA$_1$**: A generalizable embodied foundation model built upon proposed 3D-Causal world-spatial-action modeling, achieving highly competitive manipulation performance across diverse simulated and real-robot benchmarks using only 6K hours of pre-training data.

Abstract

Embodied foundation models (EFMs), which map visual observations and language instructions to continuous robotic actions, have emerged as the dominant approach for general-purpose robotic systems to date. However, current EFMs lack an innate ability to reason about physical dynamics and the causal effects of robot behaviors on the physical world. This leads to a fundamental mismatch between 2D-centric visual perception and 3D-centric embodied interaction, severely compromising the reliability and generalization of EFMs in real-world robotic tasks. To address this gap, we present **TBot-SA$_1$**, a novel embodied foundation model built upon our proposed 3D-centric World-Spatial-Action (WSA) modeling paradigm. It not only learns 3D world-aware imagination to visually plan future robotic behaviors, but also models bidirectional causal dependencies between 3D world state transitions and robotic actions for enhancing embodied behavior generalization. Notably, TBot-SA$_1$ achieves highly data-efficient pre-training with 6k hours of expert demonstration data (only 1k hours from real robot), while delivering competitive manipulation performance (93% success rate) on RobotWin2.0 simulation benchmark and achieving +20% average boosted performance on real-world robot control tasks compared to state-of-the-art EFMs.

Contents

1 Introduction

Creating versatile robots to match or exceed human-level physical intelligence has long been a core pursuit in embodied AI. Serving as the cornerstone of human physical intelligence, the innate capacity for spatial world modeling enables humans to integrate 2D visual understanding and task objectives, predict physical dynamics in 3D space, and execute precise motor actions [12, 14]. Such 2D-to-3D spatial reasoning capability is equally vital for versatile robots, as physical interactions in the real world inherently occur in 3D space and follow physical laws. Thus, developing an embodied foundation model that can master intrinsic causal relationships between spatial environment evolution and robot behaviors, along with the physical common sense underlying 3D world interactions, is the critical breakthrough needed to unlock the full potential of embodied artificial intelligence.

In recent years, multi-modal large models [11, 3, 43, 34] have achieved remarkable progress and, meanwhile, demonstrated a promising technical roadmap toward beyond human-level intelligence through training on web-scale data. Following this direction, modern embodied foundation models have gradually evolved into two typical paradigms: Vision-Language-Action (VLA) models [22, 49, 35, 30, 7, 6, 16, 47] and World Action Models (WAM) [4, 26, 9, 17, 42, 39, 15]. VLA models construct semantic-centric robotic policies by augmenting pre-trained vision-language models (VLMs) with a lightweight action expert. This design allows VLAs to inherit powerful open-world visual understanding capabilities from pre-trained VLMs, while the appended action expert generates robot actions conditioned on semantic-level visual comprehension, following an "understand-then-execute" paradigm. In contrast, WAM learns imagination-centric robot policies based on a predictive video generation model that acts as a vision planner. Based on this "imagine-then-execute" paradigm, its action expert functions as an inverse dynamics model, which generates robotic control signals corresponding to the planned visual frames.

Despite the impressive progress of current VLAs and WAMs, two fundamental challenges still impede their generalization to real-world scenarios. **First**, generative vision-language learning in VLAs and predictive video learning in WAMs are inherently 2D-centric [27, 46, 28]. Both paradigms operate primarily on 2D visual signals, failing to explicitly model the 3D physical dynamics of the environment and the bidirectional causal relationships between robot actions and scene changes. This renders them unreliable in real-world tasks that require 3D spatial reasoning and robust adaptation to environmental changes. **Second**, learning generalizable embodied foundation models heavily depends on real-world expert demonstration data, which is costly and time-consuming to acquire at scale [5]. This data scarcity further prevents models from learning the generalizable physical commonsense behind demonstrations, which is essential for developing versatile robots that can adapt to unseen scenarios [5, 4]. These challenges motivate us to study a significant question:

How to learn 3D-centric world–action causality in a shared latent space for constructing generalizable embodied foundation models under constrain of limited expert demonstration data?

In this work, we present **TBot-SA$_1$**, a novel embodied foundation model built on data-efficient World-Spatial-Action (WSA) paradigm. Specifically, the WSA breaks the inherent 2D-centric limitations of prevailing paradigms via integration of three essential capabilities. First, **at the semantic understanding level**, it enables instruction-aligned 2D visual planning via subgoal image prediction, empowering a robot to accurately parse the underlying intent of task instructions. Second, **at the causal reasoning level**, it incorporates a causal 3D scene dynamics mechanism that explicitly models the evolution of 3D scenes conditioned on robot actions, endowing a robot with the ability to capture the intrinsic causal relationships between environmental changes and robot behaviors. Third, **at the closed-loop execution level**, it learns a 3D inverse dynamics model that maps the planned 3D scene states to executable robot control signals, fully closing the loop between 3D spatial world modeling and real-world physical interaction. To this end, the TBot-SA$_1$ is implemented via a Mixture-of-Transformers (MoT) architecture with 3D-centric bidirectional causal attention, enabling efficient joint learning of all three capabilities. In summary, our main contributions are three-fold:

- **Robot Learning Paradigm:** We propose World-Spatial-Action (WSA) modeling, a novel robot learning paradigm that unifies instruction-aligned semantic understanding, causal 3D world modeling, and closed-loop physical execution into a single model, addressing the inherent 2D-centric limitations of prevailing VLA and WAM paradigms.
- **Embodied Foundation Model:** We present TBot-SA$_1$, a data-efficient embodied foundation model implementing the WSA paradigm via a Mixture-of-Transformers (MoT) architecture, enabling

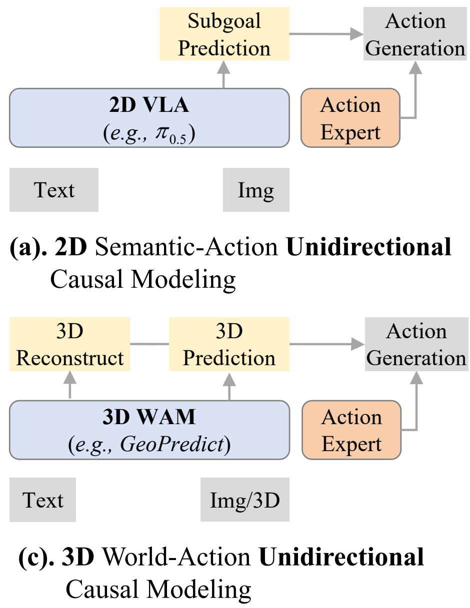


Figure 2: **Embodied Foundation Model Paradigms:** (a) 2D-centric VLA models learns unidirectional causality from visual semantics to robot action; (b) 2D-centric World Action Model learns unidirectional causality from 2D visual dynamics to robot actions; (c) 3D-centric world action model learns unidirectional causality from 3D dynamics to robot action; (d) World-Spatial-Action model unifies bidirectional causality between 3D world dynamics and robot action into a single world.

bidirectional causal reasoning between 3D world dynamics and robot actions in a shared latent space.

- **Data-efficient Learning and Strong Performance:** We pre-train TBot-SA$_1$ with 6k hours of demonstration data (only 1k hours from real robot), and it delivers competitive manipulation performance (e.g., 93% success rate on RobotWin2.0) on simulation benchmark and achieves $+20\%$ boosted performance on real-world robot control tasks.

2 Related Work

The core goal of embodied foundation models (EFMs) is to establish a robust, stable multi-modal mapping from robot observations and task instructions to executable robot actions [44, 41]. As illustrated in Figure 2, mainstream research on EFMs has converged into two dominant paradigms: Vision-Language-Action models (VLAs) and World-Action-Models (WAMs). In the following sections, we provide a systematic literature review of these two research directions.

Vision-Language-Action: As the most widely adopted paradigm for robot manipulation, VLA models [35, 7, 6, 37, 47] are generally built on pretrained vision-language models. By inheriting rich semantic priors from VLMs [11, 3], they successfully transfer open-world visual-semantic knowledge to the physical robot domain via end-to-end fine-tuning on robot demonstration data. Early representative works such as RT-2 [49] and OpenVLA [16] adopt an autoregressive strategy, which discretizes continuous robot actions and formulates policy learning as a sequence modeling task. To address the limitation of discrete tokenization in fine-grained continuous control, recent state-of-the-art works [7, 6, 5] have shifted to generative frameworks based on conditional flow matching, which directly model the continuous-valued action, significantly improving the precision and stability of robot control. In addition, several works [48, 28] augment VLA with 3D geometry priors by incorporating extra 3D information such as depth or 3D gaussian splatting, demonstrating the effectiveness of 3D representation in robot learning. However, mainstream VLA policies are essentially reactive frameworks that make decisions based only on current observations, lacking predictive planning and 3D causal reasoning capabilities, which makes them unreliable in real-world manipulation tasks that require 3D dynamics modeling.

World-Action-Model: In a different line from VLAs, WAMs follow the chain-of-thought philosophy and formulate robot policies as an imagine-then-execute procedure. Guided by this principle, numerous works [26, 39, 15, 42, 23] start with video models, inheriting rich commonsense knowledge from large-scale video generation pre-training. They then use these video models as world dynamics simulators, enhancing the reliability of robot foundation models via co-training of future frame prediction and action generation. For example, Motus [4] demonstrates that explicitly modeling future frame prediction significantly improves the generalization of robot manipulation, while Fast-WAM [42] proves that improved manipulative generalization comes from the co-learning rather than explicit video prediction. Furthermore, recent efforts [27, 45, 19] have further extended WAM paradigm to 3D space by modeling future geometry prediction. For instance, GeoPredict [27] leverages predictive kinematics and 3D Gaussian geometry to improve the manipulation performance, and PointWorld [14] scales 3D world models for in-the-wild robotic manipulation. However, existing 3D WAMs still follow a unidirectional "predict-then-execute" pipeline, failing to unify 3D world dynamics modeling and inverse dynamics into a single latent space for bidirectional causal learning. They cannot explicitly model how robot actions affect 3D space evolution and, in turn, how 3D space changes should guide action generation, which is the core focus of our work.

Compared with prior studies, our work is most closely related to WAMs but takes a further step: instead of directly co-learning video generation and robot control, our method unifies action-conditioned 3D world modeling and 3D inverse dynamics into a single model. This mechanism enables our model to exploit the intrinsic causal consistency: it not only learns how 3D robot actions effect the evolution of the 3D world but also determines the appropriate action plan given variations in the 3D world.

3 Methodology

In this section, we first formalize the robot control problem in Section 3.1. We then introduce the overall architecture of TBot-SA$_1$ in Section 3.2, followed by detailed learning objectives of the World-Spatial-Action (WSA) modeling mechanism in Section 3.3. Finally, we present the pre-train data recipe in Section 3.4.

3.1 Problem Definition and Notation

In the context of robot control, the goal of embodied foundation model is to learn a robot policy that predicts next robot actions from current observations. Formally, at each timestep t, the observations include task instruction termed l, visual observation denoted as v_t, and robot proprioception termed s_t. In general, the robot action is a H-step chunk termed $A_t := (\mathbf{a}_{t+1}, \ldots, \mathbf{a}_{t+H})$, a short-term action trajectory with a sequence of end-effector pose or joints of a robot in next H timesteps. To train a robot policy π_θ parameterized by θ on diverse robot demonstrations $\mathcal{D}_{\text{dem}}$, it is required to maximize the the standard likelihood objective:

$$\max_{\theta} \mathbb{E}_{(v_t, s_t, A_t, l) \sim \mathcal{D}_{\text{dem}}} \log \pi_\theta(A_t \mid v_t, s_t, l). \tag{1}$$

As illustrated in Figure 2, previous works formulate robot control problem as vision-language-action or world-action policy, which can be regarded as a joint distribution derived from the standard objective.

2D Vision-Language-Action Policy. The 2D-centric VLA aims to directly predict executable actions conditioned on the current 2D observation $\mathbf{O}_t := \{v_t, s_t, l\}$. It learns 2D semantic-action unidirectional causality, which is formulated as a posterior distribution:

$$\mathbf{A}_t \sim p(a_{t+1:t+H} \mid v_t, s_t, l). \tag{2}$$

2D World-Action-Model Policy. The 2D-centric WAM couples predictive visual generation with next action generation, which jointly predicts executable actions and N subgoal images $\mathbf{V}_t := (\mathbf{v}_{t+1}, \ldots, \mathbf{v}_{t+N})$ from the current observation $\mathbf{O}_t$. It learns 2D world-action unidirectional causality, which can be formulated as a joint prediction distribution:

$$\mathbf{A}_t \sim p(a_{t+1:t+H}, v_{t+1:t+N} \mid v_t, s_t, l). \tag{3}$$

3D World-Action-Model Policy. Instead of modeling 2D world-action joint distribution, the 3D-centric WAM focuses on the unidirectional causality between evolution of scenes and robot action in

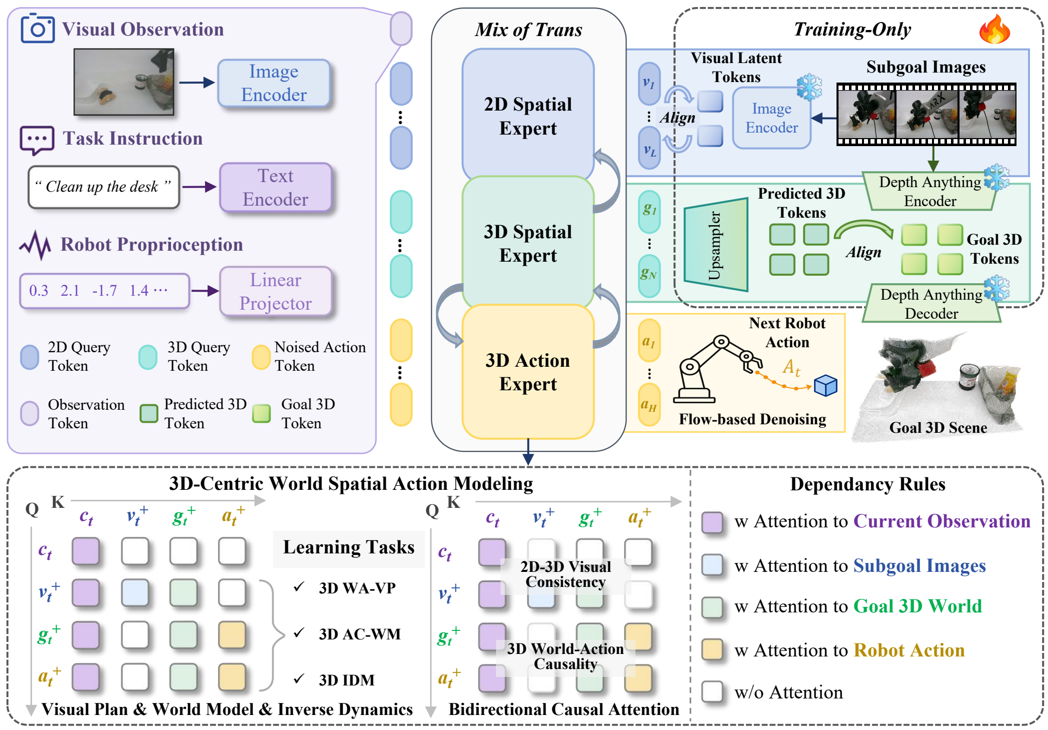


Figure 3: Overview of TBot-SA$_1$. It adopts a Mixture-of-Transformers structure with three complementary experts: a 2D Spatial Expert for instruction-aligned visual planning, a 3D Spatial Expert for action-conditioned 3D world prediction, and a 3D Action Expert for 3D robot action generation. A bidirectional causal attention mechanism enforces dependency rules to unify 2D-3D visual consistency and 3D bidirectional action–scene causality within a shared latent space, enabling joint learning of visual planning, 3D world modeling, and 3D inverse dynamics model for robot control.

3D space. Specifically, given 2D observation $\mathbf{O}_t$ and 3D observation g_t, it learns to predict K subgoal 3D scenes $\mathbf{G}_t := (\mathbf{g}_{t+1}, \ldots, \mathbf{g}_{t+K})$ for robot control. Formally, this models a joint prediction distribution as followed:

$$\mathbf{A}_t \sim p(a_{t+1:t+H}, g_{t+1:t+K} \mid g_t, v_t, s_t, l). \tag{4}$$

3D World Spatial Action Policy. Different from previous works, we formulate robot control as 3D-centric world-spatial-action learning, which jointly models 2D semantic-world unidirectional causality and 3D world-action bidirectional causality. Given the current observations $\mathbf{O}_t$, the WSA models a joint distribution upon three prediction tasks that are relevant to robot control, including 3D-conditioned subgoal image prediction, action-conditioned 3D scene prediction and 3D-conditioned action generation:

$$\mathbf{A}_t \sim p(\mathbf{V}_t, \mathbf{G}_t, \mathbf{A}_t \mid \mathbf{O}_t) = p(\mathbf{V}_t \mid \mathbf{O}_t, \mathbf{G}_t) \cdot p(\mathbf{G}_t \mid \mathbf{O}_t, \mathbf{A}_t) \cdot p(\mathbf{A}_t \mid \mathbf{O}_t, \mathbf{G}_t). \tag{5}$$

3.2 Model Architecture

Inspired by the recent works [4, 6], TBot-SA$_1$ is built upon a unified multi-modal transformer, which leverages mixture-of-transformer (MoT) architecture to enable seamless integration of 2D visual generation, future 3D prediction and robot action generation. As illustrated in Figure 3, it involves three major components: 2D spatial expert, 3D spatial expert and 3D action expert.

2D Spatial Expert (2D-SE): Given a human instruction, generating a sequence of visual changes in 2D space enhances understanding of human intent. The 2D-SE $f_{2D}(\cdot)$ serves as the instruction-aligned semantic understanding branch of TBot-SA$_1$. It is initialized from a pre-trained vision-language model such as QWen3-VL [2] or Wan2.1 [31], thereby inheriting strong vision-language semantic priors. The goal of 2D-SE is to predict future dynamics that describe how 2D visual appearance changes so that follow the instruction. Specifically, it predicts highly abstracted tokens $h_v = f_{2D}(O_t, G_t)$ for subgoal images, instead of RGB pixels. This design enhance visual understanding capability of the 2D-SE, as it is required to focus on essential features of an image rather than redundant pixel information.

3D Spatial Expert (3D-SE): Given a sequence of executable robot actions, predicting evolution of world dynamics in 3D space reflects causal effect of the robot behavior. Under this context, the 3D-SE $f_{3D}(\cdot)$ conducts a internal mechanism for 3D causal world modeling. To capture essential feature of 3D scenes, the 3D-SE predicts latent representations $h_g = f_{3D}(O_t, A_t)$ for subgoal 3D scenes. We construct it with a transformer-based module of the same depth as 2D-SE, thereby the architecture of 3D-SE can be directly derived from that of 2D-SE.

3D Action Expert (3D-AE): Given a demonstration of 3D space evolution, predicting next robot action tells why and how robot to behave in 3D space. The 3D-AE $f_{act}(\cdot)$ aims to predict a full action chunk conditioned on both current observations and future 3D latent representations. Specifically, we instantiate 3D-AE as denoising diffusion transformer. Given noisy actions $\hat{A}_t = \tau A_t + (1-\tau)\omega$, where $\omega \sim \mathcal{N}(0,1)$ and $\tau \in [0,1]$, it employs iterative cross-attention and denoising procedure to predict latent action representations $h_{act} = f_{act}(O_t, G_t, \hat{A}_t)$. This yields a 3D inverse dynamics formulation, where 3D-AE maps predicted 3D latent evolution back to executable action.

3D-centric World-Action Causal Attention: Three experts are coupled within the mixture-of-transformer architecture via a shared attention mechanism. Formally, let h_t denote tokens of current observation, which are extracted via 2D-SE. As illustrated in Figure 3, we adopt a 3D-centric causal attention mechanism, which defines the following dependency rules: (1) All predicted tokens (h_v, h_g, h_{act}) can attend to the current observation tokens h_t, ensuring all predictions are grounded in the current state of the robot and the environment. (2) The visual tokens of future subgoal images h_v can attend to the 3D scene tokens h_g, encouraging the model to learn 2D-3D visual consistency and ground semantic visual predictions in physically consistent 3D geometry. (3) The 3D scene tokens h_g and action tokens h_{act} have full bidirectional attention access to each other, enabling joint learning of action-conditioned 3D dynamics prediction and 3D scene-aware action generation. This bidirectional attention is the key to bidirectional causal modeling of the WSA paradigm.

3.3 Learning Objective

TBot-SA$_1$ is optimized end-to-end with three joint training objectives, which correspond to the three factorized terms in the WSA joint distribution (Equ. 5) and align with the three core capabilities of the WSA paradigm:

3D World-Aware Visual Planning (3D WA-VP). We adopt a MSE loss over the predicted visual tokens, to minimize the discrepancy between the predicted subgoal image tokens and the ground-truth tokens extracted via pre-trained tokenizer. Formally, the loss is defined as:

$$\mathcal{L}_{\text{2D}} = \mathbb{E}_{(O_t, G_t, V_t) \sim \mathcal{D}_{dem}} \left\| f_{2D}(O_t, G_t) - f_{enc}(V_t) \right\|_2^2, \tag{6}$$

where $f_{enc}(\cdot)$ denotes a pre-trained VAE-based tokenizer (e.g., COSMOS [1]).

Action-Conditioned 3D World Modeling (3D AC-WM). We adopt MSE loss over the predicted 3D scene latent representations, to minimize the error between the predicted 3D scene evolution and the ground-truth 3D geometric states from the demonstration data. The loss is formalized as:

$$\mathcal{L}_{\text{3D}} = \mathbb{E}_{(O_t, A_t, G_t) \sim \mathcal{D}_{dem}} \left\| f_{3D}(O_t, A_t) - f_g(G_t) \right\|_2^2, \tag{7}$$

where $f_g(\cdot)$ denotes a pre-trained 3D foundation model, e.g., Depth-Anything [20].

3D Inverse Dynamics Modeling (3D IDM). Following previous works [27, 6], we adopt flow matching loss [21] for continuous action generation. This loss minimizes the error between the predicted flow field and the ground-truth one, conditioned on the current observation and the 3D scene state evolution. The loss is defined as:

$$\mathcal{L}_{\text{ACT}} = \mathbb{E}_{(O_t, G_t, A_t) \sim \mathcal{D}_{dem}} \left\| \omega - A_t - f_{act}(O_t, G_t, \hat{A}_t) \right\|_2^2. \tag{8}$$

Table 1: Data mixture used for pretraining.

Data source	Type	Num. frames	Num. tasks	Sampling weight
InternData-A1 [32]	Sim.	396M	70	0.64
RoboTwin [10]	Sim.	17M	50	0.08
AgiBot-World [8]	Real	206M	217	0.18
RoboChallenge [36]	Real	5M	30	0.02
EgoDex [13]	Human	68M	194	0.08

This trains 3D-AE to predict flow field ($\omega - A_t$), thereby facilitating coherent action generation. The overall learning objective is the sum of the three losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{2D}} + \mathcal{L}_{\text{3D}} + \mathcal{L}_{\text{ACT}}. \tag{9}$$

We adopt two-stage strategy to optimize the TBot-SA$_1$: pre-training on diverse data sources to acquire real-world behavior priors in first stage; post-training on specific manipulation tasks to adapt learned priors to target embodiment in second stage. Different from previous works [9, 4], TBot-SA$_1$ is trained by co-learning of three capabilities in both pre-training and post-training stage, rather than using a hierarchical manner.

3.4 Pre-training Data

High-quality embodied data is the cornerstone of pre-training generalizable embodied foundation models that can acquire versatile manipulation skills for flexible real-world robot control. However, due to the prohibitive cost and safety constraints of human-robot teleoperation, collecting large-scale high-quality real-world robot data at scale remains challenging [5]. To address this limitation, alternative embodied data sources such as simulation data and egocentirc human data have emerged as critical complements. This gives rise to a multi-sourced data pyramid [5, 4], where data quantity decreases from the bottom to the top. It suggests that each data layer exhibits distinct, complementary strengths and inherent limitations, and no single data source alone can support the development of generalizable embodied foundation model: human data (at the bottom of the pyramid) provides the "what" and "why" of tasks, simulation robot data (at the middle) offers the "how" for robot control, and real-world robot data (at the top) bridges the "reality gap" between simulation and physical environments [40]. Their synergistic integration is therefore not merely beneficial but essential for pre-training robust, versatile, and human-aligned robot foundation models that can generalize across tasks, platforms, and physical environments. Accordingly, as illustrated in Table1, we pre-train the model on a mixture of heterogeneous data sources:

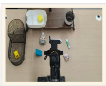
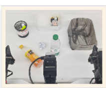
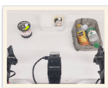
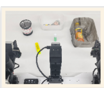
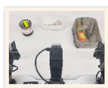
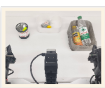
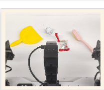
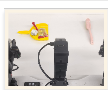
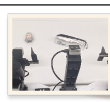
- **Simulation data:** We collect simulated synthetic data are from InternData-A1 [32] and RoboTwin2.0 [10], providing diverse and low-cost manipulation trajectories for learning scene dynamics, object interaction, and coordinated control.
- **Human data:** We include egocentric human manipulation videos from EgoDex [13], which provide rich hand-object interaction priors and further enhance the model's understanding of manipulation behaviors and scene evolution.
- **Real-world data:** The real-world robot data are drawn from RoboChallenge [36] and AgiBot-World [8], which help reduce the sim-to-real gap and improve robustness in physical environments.

The mixture of data sources covers 10 different robot embodiments and 200 robot control tasks, including 60 atomic manipulation skills such as grasping and placing. Notably, although the dataset provides sufficient diversity, it is not particularly large in scale: the total duration of all manipulation videos is only over 6,000 hours.

4 Experiments

In this section, we introduce comprehensive evaluation experiments, which are conducted to explore the following questions: 1) How does proposed TBot-SA$_1$ perform compared to existing

Table 2: Real-robot experimental setup and task details.

Initial	Finished	Robotic Embodiment	Task Description
		Unimanual: AgileX PiPER (7-DoF)	**RGB Block Sorting**: Position red block, green block, and blue block from left to right in the specified sequence.
		Unimanual: AgileX PiPER (7-DoF)	**Object Organization**: Tidy up the table by sorting the toys, stationery, and trash into their designated containers.
		Bimanual: ARX Lift2 (14-DoF)	**Trash Cleaning**: Pick up the trash with the nearest gripper, hand it over to the other gripper, and place it into the trash bin.
		Bimanual: ARX Lift2 (14-DoF)	**Pen Holder Placement**: Pick up the markpen and place it into the pen holder.
		Bimanual: ARX Lift2 (14-DoF)	**Toy Box Organization**: Pick up the toys from the desktop, place them into the toy box, and arrange them neatly inside.
		Bimanual: ARX Lift2 (14-DoF)	**Sweep Trash**: Sweep the trash into the dustpan using a broom.
		Bimanual: ARX Lift2 (14-DoF)	**Unzip Pencil Bag**: Grasp the pencil bag, unzip it, and place it back onto the tabletop.

VLA/WAM models across multiple benchmarks and emobodiments? 2) How does the design choice in proposed WSA paradigm contribute to open-world robot learning performance? Next, we first present experimental setup. Then, we demonstrate the merit of TBot-SA$_1$ through a comparison with state-of-the-arts on both simulation and real-world tasks. Finally, we conduct ablation studies to illustrate major properties of proposed WSA paradigm.

4.1 Experimental Setup

Real-robot setup Real robot experiments are conducted with two embodiments: (1) Unimanual AgileX PiPER with three cameras and 6-DoF robotic arm; (2) Bimanual ARX Lift2 with three cameras and two 7-DoF robotic arms. To ensure diversity of robot behaviors in real-world scenario and investigate cross-embodiment generalization, we design 7 tabletop manipulation tasks. For each task, objects are randomly initialized within the workspace to evaluate model's generalization capability across object position, orientation, and scene layout. The robot receives language instructions and visual observations as input, and predicts continuous action chunks for closed-loop execution. For testing, we adopt two metrics for comprehensive evaluation of EFMs: (1) Success Rate (SR), the proportion of rollouts in which the task is successfully completed. (2) Completeness (C), the proportion of rollouts in which the sub-task is completed.

Simulation setup We adopt RoboTwin-2.0 benchmark [10] for evaluation in simulation environment. RoboTwin platform is setup with easy and hard modes, where the former offers clean environment without any distracted factor and the latter conduct domain randomization with background changes and distracted objects. Following official protocol [10], 27500 demonstrations (2500 for clean, 25000 for hard) are adopted for model fine-tuning. In addition, we adopt multi-task test setting (i.e., train for once, test for all tasks) to evaluate how much versatility of each model attain. During testing, success rate is adopted as the evaluation metric following the official protocol.

Table 3: Real-world evaluation results on seven tabletop manipulation tasks. **SR** denotes the task success rate (%), and **C** denotes the task progress score (%).

Method		π_0		$\pi_{0.5}$		InternVLA-A1		TBot-SA$_1$	
Model Type & Model Size		**VLA**	**3B**	**VLA**	**3B**	**WAM**	**3B**	**WSA**	**3B**
Evaluation Results on 7 Real-world Tabletop Robot Manipulation Tasks									
Task Name	**Robot Type**	**SR**	**C**	**SR**	**C**	**SR**	**C**	**SR**	**C**
RGB Block Sorting	AgileX Piper	70	70	70	70	50	50	**80**	**80**
Object Organization	AgileX Piper	50	50	77	77	37	37	**80**	**80**
Trash Cleaning	ARX Lift2	37	53	77	87	43	58	**80**	**90**
Pen Holder Placement	ARX Lift2	7	28	70	77	33	38	**87**	**90**
Toy Box Organization	ARX Lift2	13	18	13	27	37	48	**60**	**72**
Sweep Trash	ARX Lift2	30	40	27	44	44	78	**80**	**89**
Unzip Pencil Bag	ARX Lift2	30	41	55	63	25	43	**75**	**75**
Average (%)		34	36	55	59	38	53	**77**	**83**

4.2 Main Results on Real-World Tasks

Real-world Evaluation Tasks As illustrated in Table 2, the real-world benchmark consists of 7 representative tabletop scenarios: RGB Block Sorting, Object Organization, Trash Cleaning, Pen Holder Placement, Toy Box Organization, Sweep Trash and Unzip Pencil Bag. These tasks require the robot to perceive cluttered scenes, interact with objects of different categories and spatial layouts, and complete multi-step manipulation behaviors such as grasping, placing, pushing, and sweeping. Compared with simple pick-and-place tasks, these scenarios place higher demands on spatial reasoning, long-horizon execution, and robustness to diverse object arrangements.

Baselines We compare TBot-SA$_1$ with three representative and open-source embodied foundation models: π_0, a generalist policy for continuous action prediction; $\pi_{0.5}$, a policy for open-world robot control that demonstrates impressive generalization; and InternVLA-A1, a unified model integrating visual understanding, video generation and action generation. For evaluation, we report the mean task success rate and the task progress score. The success rate measures whether the whole task is completed successfully, while the progress score reveals progress of full task execution.

Evaluation Results As shown in Table 3, TBot-SA$_1$ achieves leading performance over baselines under real-world setting. Across 7 tabletop manipulation tasks, TBot-SA$_1$ achieves an average success rate of 77% and an average completeness score of 83%, substantially outperforming the second-best model $\pi_{0.5}$, which obtains 55% and 59%, respectively. Moreover, the improvement is consistent across both the single-arm PiPER and bimanual Lift2 platforms, with TBot-SA$_1$ achieving the highest success rate on each task. These results highlight the effectiveness of the proposed WSA architecture. Rather than relying only on reactive 2D visual-action mapping, TBot-SA$_1$ couples instruction-aligned visual prediction, 3D world dynamics, and 3D-conditioned action generation, leading to more reliable execution in spatially complex and long-horizon tasks. Notably, this performance is achieved with only moderate-scale pre-training data, suggesting that WSA provides an effective inductive bias for learning transferable real-world manipulation skills under limited demonstration scale.

4.3 Main Results on Simulation Benchmark

Simulation Tasks RoboTwin2.0 [25, 10] is a challenging dual-arm manipulation benchmark that evaluates policy generalization under both clean and randomized environments. It covers diverse bimanual skills, including object handover, coordinated manipulation, spatially constrained interaction, and long-horizon tabletop tasks.

Evaluation Results As shown in Table 4, TBot-SA$_1$ achieves the best performance across open-source models, obtaining a success rate of 93% in the hard setting. Compared with large WAM models such as *Motus* and *Linbot-VA* (model size $> 3B$), TBot-SA$_1$ with a model size of 3B delivers higher performance, and meanwhile remains competitive with the best closed-source model

Model Name	**Access Status**	Model Size	Model Type	**SR** (%)
Qwen-VLA	Closed-source	5B	VLA	87
Being-H0.7	Closed-source	3B	WAM	90
MotuBrain	Closed-source	Unknown	WAM	96
π_0	Open-source	3B	VLA	58
$\pi_{0.5}$	Open-source	3B	VLA	77
ABot-M0	Open-source	4B	VLA	85
Motus	Open-source	8B	WAM	87
InternVLA-A1	Open-source	3B	WAM	88
LingBot-VA	Open-source	7B	WAM	92
Fast-WAM	Open-source	6B	WAM	92
TBot-SA$_1$(Ours)	Open-source	3B	WSA	**93**

Table 4: RoboTwin2.0 benchmark experimental results. TBot-SA$_1$ is evaluated against closed-source and open-source embodied foundation models under hard setting. Mean success rate **SR** over 50 manipulation tasks is reported. TBot-SA$_1$ achieves leading performance over open-sourced models.

Model Name	**Access Status**	Model Size	Model Type	**SR** (%)
Qwen-VLA [33]	Closed-source	5B	VLA	87
Being-H0.7 [24]	Closed-source	3B	WAM	90
MotuBrain [29]	Closed-source	Unknown	WAM	96
π_0 [7]	Open-source	3B	VLA	58
$\pi_{0.5}$ [6]	Open-source	3B	VLA	77
ABot-M0 [38]	Open-source	4B	VLA	85
Motus [4]	Open-source	8B	WAM	87
InternVLA-A1 [9]	Open-source	3B	WAM	88
LingBot-VA [18]	Open-source	7B	WAM	92
Fast-WAM [42]	Open-source	6B	WAM	92
TBot-SA$_1$(Ours)	Open-source	3B	WSA	**93**

MotuBrain. As illustrated in Table 5, TBot-SA$_1$ exhibits generalizable capability on long-horizon and spatially demanding tasks such as *Move Pillbottle Pad*, *Stack Blocks Three*, and *Open Microwave*. These results suggest that the proposed WSA paradigm is beneficial for learning transferable 3D world-action priors that support robust bimanual manipulation under diverse conditions.

4.4 Ablation Study

In this section, we seek to answer the second question via two ablative experiments. First, we perform ablation studies to identify the contribution of causal modeling in WSA, which involves three learning tasks: visual planning, 3D world prediction and 3D action generation. Second, we study the effect of pre-training to reveal the details of full training process in WSA.

Investigation of WSA modeling We adopt the π-style VLA policy proposed in prior works [7, 6] as the baseline model, which integrates the pre-trained QWen3-VL 3B vision-language model [2] with a randomly initialized diffusion transformer serving as the action expert. All ablation experiments are conducted on the RoboTwin2.0 platform, and the corresponding results are presented in Figure 4. Specifically, the baseline model achieves a success rate (SR) of 59.8%. Notably, learning visual planning improves the SR of action generation to 75.7%, while learning 3D world model yields a competitive SR of 74.7%. Full WSA modeling framework further outperforms all individual variants, reaching the best SR of 79.5%. Besides, modeling visual planning is beneficial for manipulation tasks that rely heavily on visual semantics, object appearance, and temporally observable state changes, such as picking bottles and open microwave. In contrast, learning 3D world model improves contact-rich manipulation tasks such as rotate QR Code and stack blocks, which require a deep understanding

Table 5: Comparison with state-of-the-art open-source models on RoboTwin2.0 under clean and randomized settings. Per-task success rate is reported.

Method	$\pi_{0.5}$ [6]		InternVLA-A1 [9]		Motus [4]		FastWAM [42]		TBot-SA$_1$	
Type & Size	**VLA**	**3B**	**WAM**	**3B**	**WAM**	**8B**	**WAM**	**6B**	**WSA**	**3B**
Evaluation Results on 50 Simulated Bimanual Robot Manipulation Tasks										
Task Name	**Clean**	**Rand.**	**Clean**	**Rand.**	**Clean**	**Rand.**	**Clean**	**Rand.**	**Clean**	**Rand.**
Place Dual Shoes	75	75	91	88	93	87	94	88	95	94
Move Stapler Pad	56	42	52	67	83	85	77	64	65	75
Stack Blocks Two	97	100	100	100	100	98	100	100	100	100
Scan Object	72	65	86	83	67	66	89	92	85	88
Place Object Stand	91	85	89	88	98	97	90	94	95	95
Place Fan	87	85	98	98	91	87	96	96	98	99
Move Pillbottle Pad	84	61	98	99	93	96	100	99	99	100
Pick Dual Bottles	93	63	92	87	96	90	100	96	94	89
Blocks Ranking Rgb	92	85	92	92	99	97	100	100	95	95
...... *(50 tasks)*										
Turn Switch	62	54	56	62	84	78	61	59	61	75
Pick Diverse Bottles	81	71	77	77	90	91	80	85	86	76
Place Bread Basket	77	64	90	91	91	94	91	93	99	94
Stack Blocks Three	91	76	90	93	91	95	95	97	98	98
Put Bottles Dustbin	84	79	90	93	81	79	95	90	95	92
Place Can Basket	62	62	81	72	81	76	71	69	80	79
Stamp Seal	79	55	75	75	93	92	90	94	73	82
Hanging Mug	18	17	31	47	38	38	58	62	50	50
Handover Block	66	57	90	73	86	73	95	81	91	84
Stack Bowls Three	77	71	89	90	79	87	80	81	92	93
Place Object Basket	80	76	89	87	81	87	89	88	84	87
Open Microwave	34	77	100	99	95	91	62	45	100	100
Average (%)	83	77	89	88	89	87	92	92	**92**	**93**

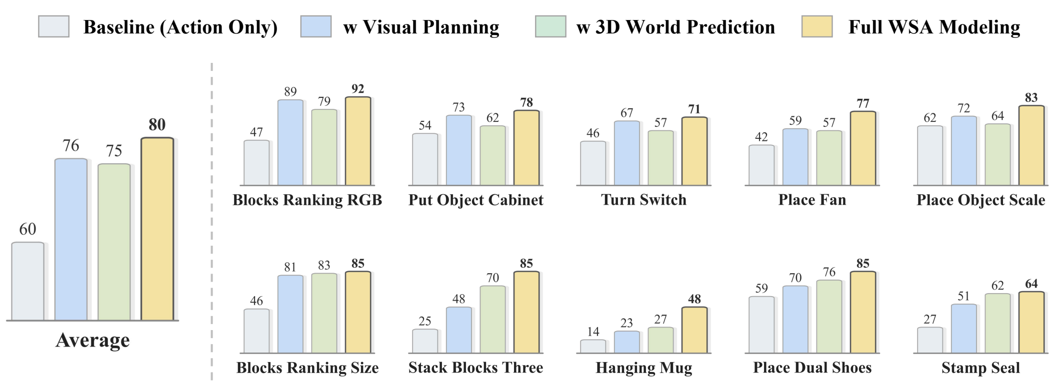


Figure 4: Investigation of WSA modeling on RoboTwin2.0 Benchmark.

of spatial geometry, object pose, and hand-object interaction. Based on the above observations, two key conclusions can be drawn. First, learning world dynamics in both 2D and 3D latent space is essential for generating generalized robot actions. Second, the bidirectional 3D world-action causal mechanism imposes effective constraints on action generation, thereby significantly enhancing the robustness of robot action learning.

The effect of pre-training To comprehensively investigate the efficacy of WSA pre-training, we design three comparative ablation settings: (1) WSA modeling is only applied in the post-training stage without pre-training on diverse heterogeneous robot datasets; (2) WSA modeling is adopted in the pre-training stage, while the post-training process only optimizes the action generation objective; (3) the model is consistently optimized with WSA modeling in both pre-training and post-training stages. As summarized in Figure 5, WSA pre-training delivers a substantial performance gain, increasing the SR from 79.5% to 88.5%. Moreover, applying WSA modeling in both pre-training

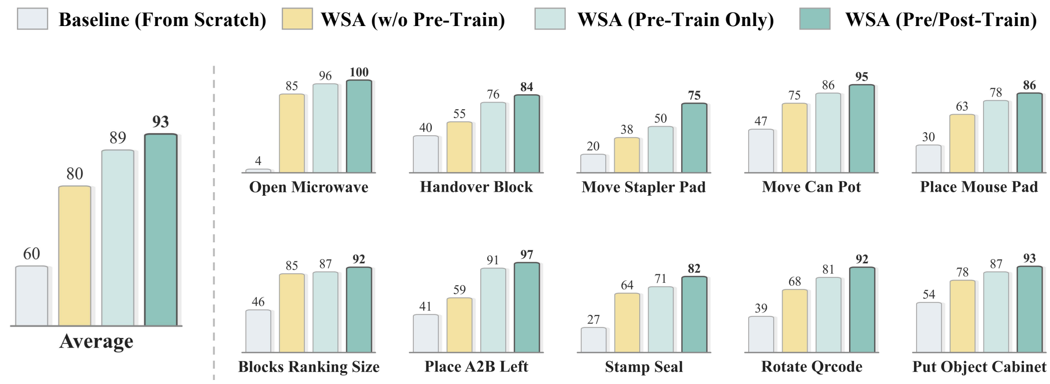


Figure 5: Ablation studies of WSA pre-training on RoboTwin2.0 Benchmark.

and post-training further boosts the SR to 91.4%. These results demonstrate that WSA serves as an effective and essential training paradigm for building robust and generalizable embodied foundation models.

4.5 Qualitative Analysis

To reveal how TBot-SA$_1$ performs robot control, we present qualitative visualizations of the model's predictions on various tasks. As illustrated in Figure 6, the visualization covers four views: visual observation with text instruction, predicted visual plan, goal 3D world prediction with 3D Gaussian and depth map, and trajectory of executed actions. From the results, we find that predicted visual plans well capture the task objective and well align the generated 3D world. Furthermore, driven by explicit 3D world modeling, the action trajectories are well-aligned with the predicted 3D goal states. These visual results demonstrate that TBot-SA$_1$ unifies semantic visual planning, causal 3D world dynamics, and closed-loop action generation within a shared latent space. It explicitly infers how actions alter the 3D world and how 3D world states guide action generation, thus leading to robust and generalizable manipulation behaviors.

5 Conclusion

This work addresses critical limitations of current embodied foundation models paradigms, which lack understanding of 3D physical dynamics and bidirectional world–action causality. We propose the World-Spatial-Action (WSA) paradigm and introduce TBot-SA$_1$, a data-efficient embodied foundation model for generalizable robot control. Built on Mixture-of-Transformers architecture and 3D-centric bidirectional causal mechanism, proposed TBot-SA$_1$ jointly learns instruction-aligned visual planning, action-conditioned 3D world modeling, and 3D-aware action generation in a unified latent space. The model is pre-trained with only 6,000 hours of demonstration data (1,000 hours from real robots), yet achieves strong performance across simulation and real-world benchmarks. Our work demonstrates that modeling 3D world–action bidirectional causality enables robots to inherently acquire physical commonsense and causal mechanisms, paving a new path for data-efficient and generalizable robot control.

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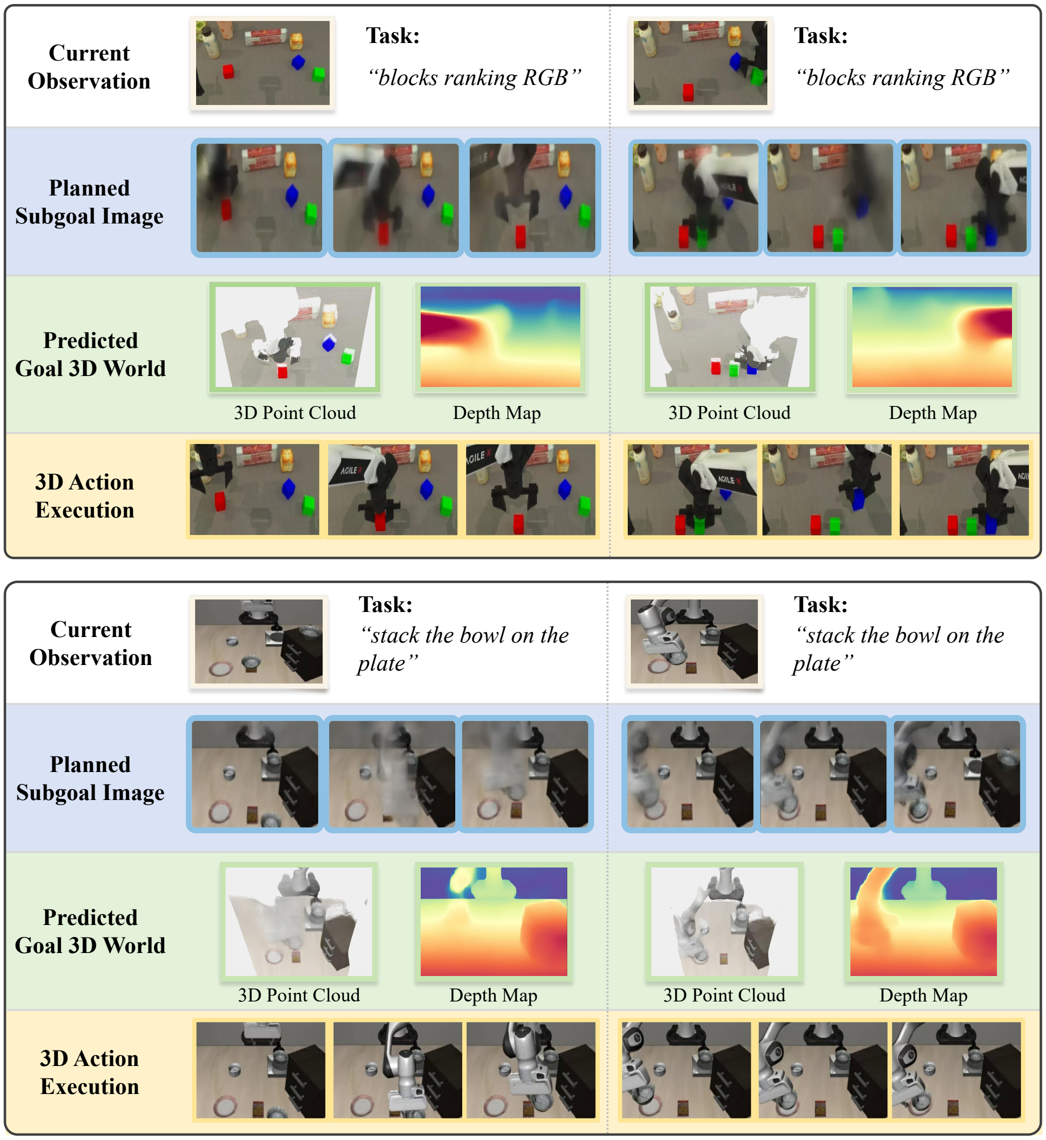

Figure 6: Visualization examples generated from TBot-SA_1. First row: the current observations with text instruction. Second row: generated subgoal images for visual planning. Third row: predicted 3D world that is rendered via Depth Anything model [20]. Fourth row: the real trajectory of executed actions.

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Task Name	**Clean**	**Random**
adjust_bottle	100	99
beat_block_hammer	98	95
blocks_ranking_rgb	95	95
blocks_ranking_size	92	92
click_alarmclock	100	100
click_bell	100	100
dump_bin_bigbin	97	99
grab_roller	100	100
handover_block	91	84
handover_mic	100	100
hanging_mug	50	50
lift_pot	100	99
move_can_pot	86	95
move_pillbottle_pad	99	100
move_playingcard_away	100	100
move_stapler_pad	65	75
open_laptop	100	100
open_microwave	100	100
pick_diverse_bottles	86	76
pick_dual_bottles	94	89
place_a2b_left	92	97
place_a2b_right	94	94
place_bread_basket	99	94
place_bread_skillet	90	89
place_burger_fries	100	100
place_can_basket	80	79
place_cans_plasticbox	100	99
place_container_plate	99	100
place_dual_shoes	95	94
place_empty_cup	100	99
place_fan	98	99
place_mouse_pad	86	86
place_object_basket	84	87
place_object_scale	85	88
place_object_stand	95	95
place_phone_stand	99	97
place_shoe	99	99
press_stapler	83	78
put_bottles_dustbin	95	92
put_object_cabinet	90	93
rotate_qrcode	87	92
scan_object	85	88
shake_bottle	100	100
shake_bottle_horizontally	100	100
stack_blocks_three	98	98
stack_blocks_two	100	100
stack_bowls_three	92	93
stack_bowls_two	97	100
stamp_seal	73	82
turn_switch	61	75
AVG.	92	93

Table 6: Per-task success rates on RoboTwin2.0 under clean and randomized evaluation settings.